



Lessons

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Median and Mode

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Toolbox

Median and Mode

There are other typical values that might be more representative of a set of data. The median is the middle value of a set of data. To find the median, sort the data from least to greatest. For an odd number of values, the middle number is the median. For an even number of values, the median is the mean of the two middle numbers.

Example 1

Hide

Angela has helped 5 dogs onto the examination table. Their weights are 40 lb, 10 lb, 140 lb, 35 lb, and 30 lb. What is the median weight of the dogs?

Step 1
Sort the numbers numerically from least to greatest.

10 lb 30 lb 35 lb 40 lb 140 lb

Step 2
Choose the middle number from the set. (If there is an even number of values, find the mean of the two middle values.)
The median weight is 35 lb.

Another typical value is the mode—the value in the set that occurs most often. (If each value occurs only once, there is no mode in the set.)

Example 2

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A restaurant offers these prices on complete meals: Fried Chicken—\$8.99, Steak—\$10.99, Fish—\$8.99, Shrimp—\$10.99, Lasagna—\$8.99, Hamburger—\$7.99. What is the mode for these prices?
Count how many times each value occurs.

\$7.99: 1 time \$8.99: 3 times \$10.99: 2 times

Because it occurs the most (3 times), \$8.99 is the mode.

Sometimes, one or more data values are very large or very small compared to the other values in the set. These extreme values are called outliers. Outliers don’t affect the median as much as they affect the mean.

Example 3

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A jewelry store has 5 different rings on sale for \$27, \$26, \$33, \$31, and \$108. Your common sense tells you that \$30 is a typical price for a ring. Compare how the outlier of \$108 impacts the mean more than the median.

Find the mean of the 5 rings:
 $\$27 + \$26 + \$33 + \$31 + \$108 = \225
 $\$225 \div 5 = \underline{\$45}$
The mean, \$45, is much higher than \$30.
Find the median of the 5 rings:
\$26 \$27 \$31 \$33 \$108
The middle value is \$31.
The median, \$31, is very close to \$30.

Notice in the example above that while the mean may not be a typical value, the median is.

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Lesson 3: Averages

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